

Basic Principles of Medicine 1

Module : Foundations to Medicine

Directed Learning activity

Title: Structure and function-Lipids: Part 2

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Complex lipids: Phospholipids and Glycolipids (Part 2)

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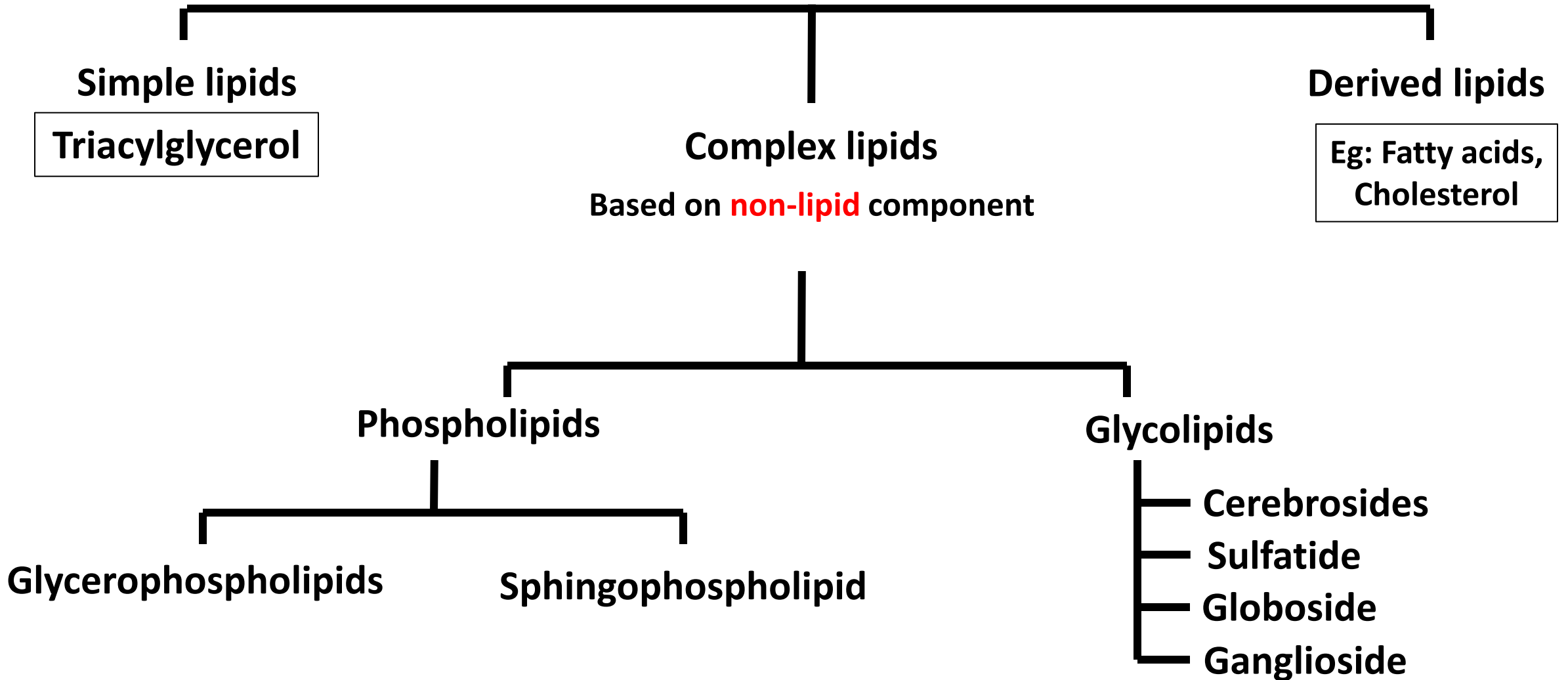
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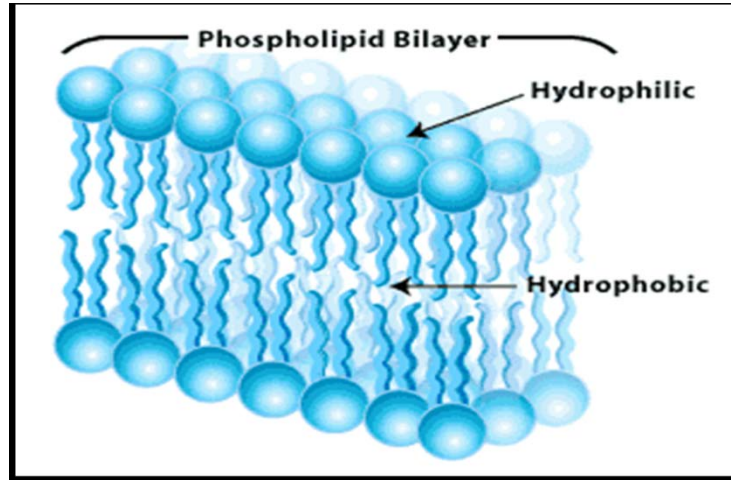
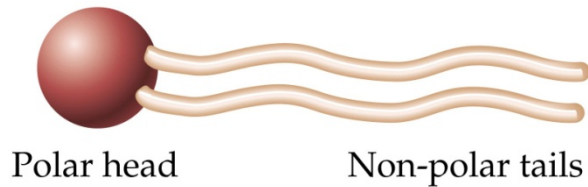
Complex lipids: Phospholipids and Glycolipids (Part 2)

SOM.MK.I.BPM1.1.FTM.3.BCHM.0007	Distinguish between phospholipids (Sphingophospholipid and glycerophospholipids) and glycolipids with examples for each
SOM.MK.I.BPM1.1.FTM.3.BCHM.0008	Indicate composition and functions of: Glycerophospholipids (Phosphatidylcholine, phosphatidylethanolamine, phosphatidylserine, phosphatidylinositol, cardiolipin), Sphingophospholipid (Sphingomyelin), Glycolipids (Cerebrosides, globosides and gangliosides)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0009	Discuss role of lung surfactant (Dipalmitoyl phosphatidylcholine/ DPPC) in respiratory distress syndrome

Classification of lipids

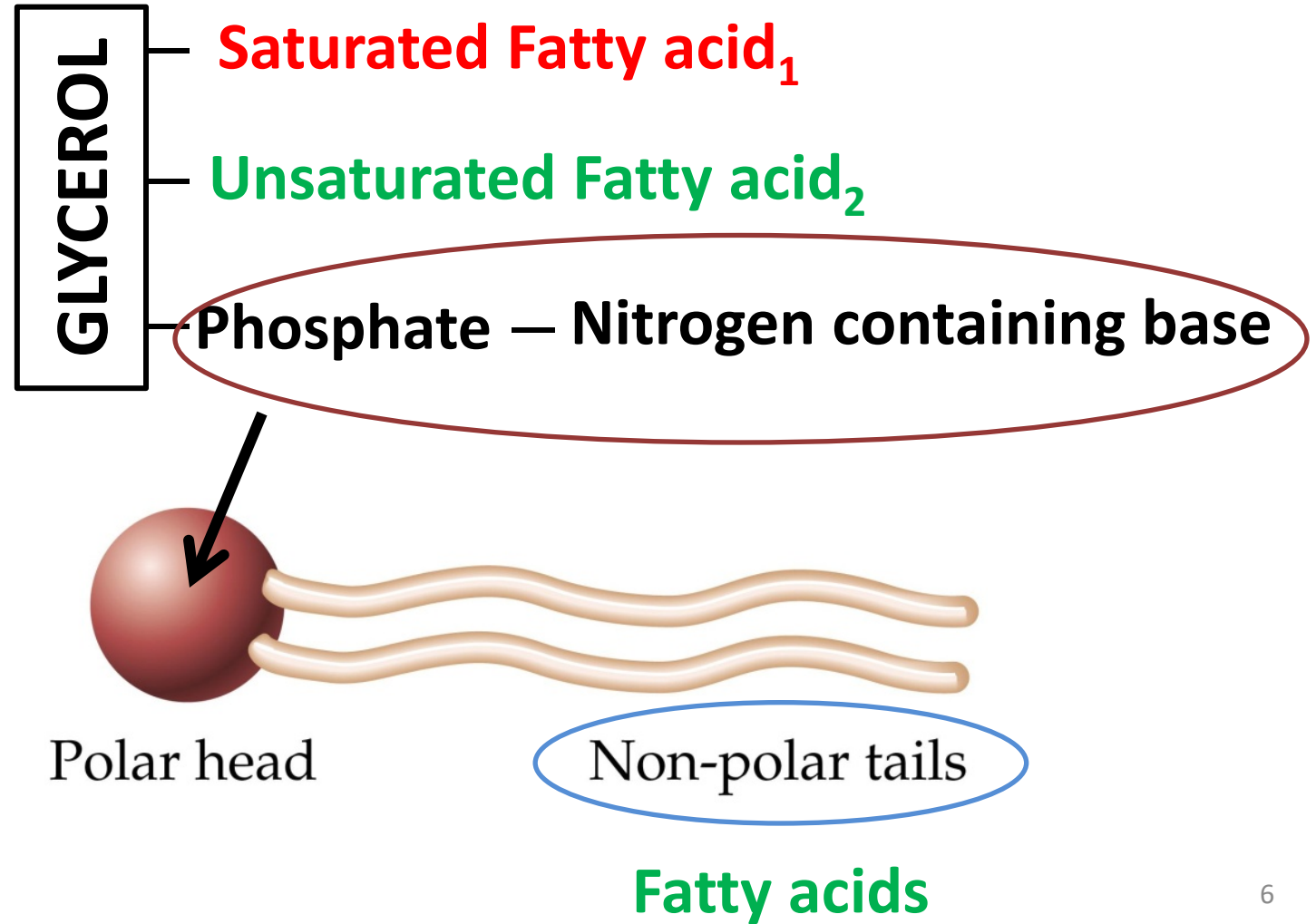
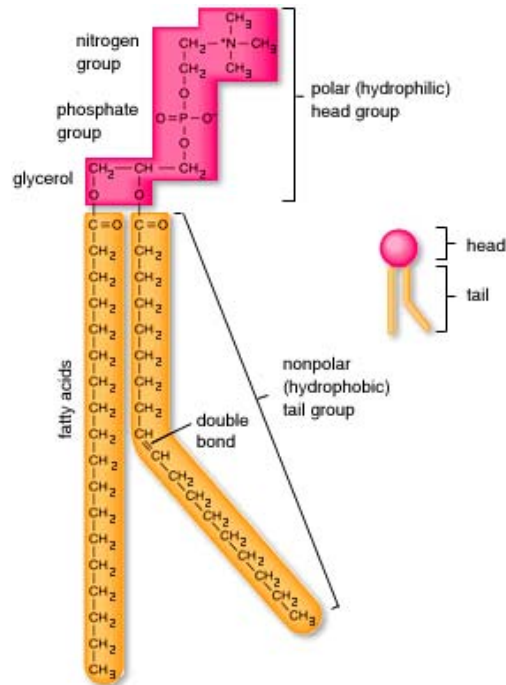


Complex lipids: Amphipathic

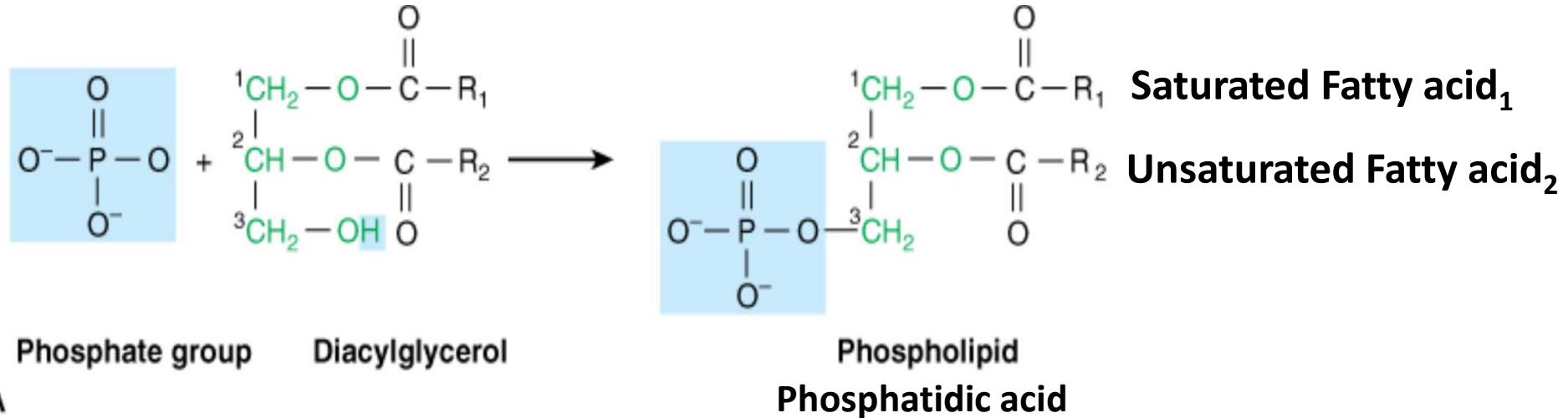


- **Amphipathic nature**
- **Unsaturated vs Saturated** fatty acid
- Unsaturated fatty acid susceptible to free radical damage (reactive oxygen species)

Glycerophospholipids: Composition



Phosphatidic acid

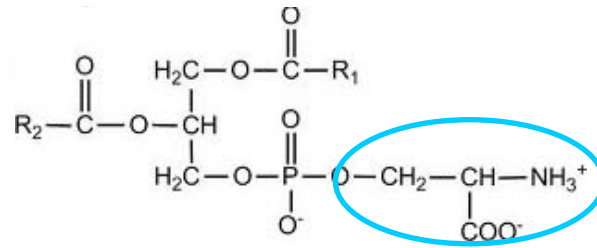


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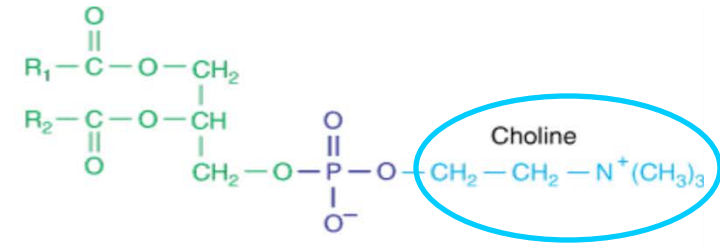
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Phosphatidic acid: Bonding between hydroxyl (OH) of phosphate group with hydroxyl (OH) of glycerol carbon 3 (green) results in Phosphatidic acid. Phosphatidic acid is **building block** of phospholipids found in cell membrane. Fatty acid chains are depicted in **black**.

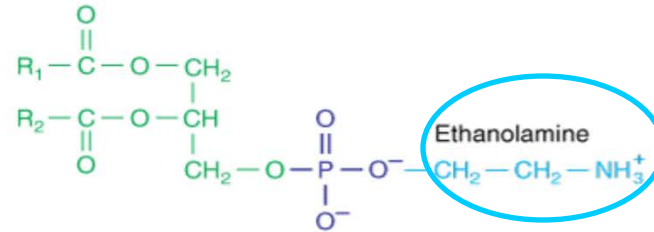
(Phosphatidic acid is NOT found in cell membranes)



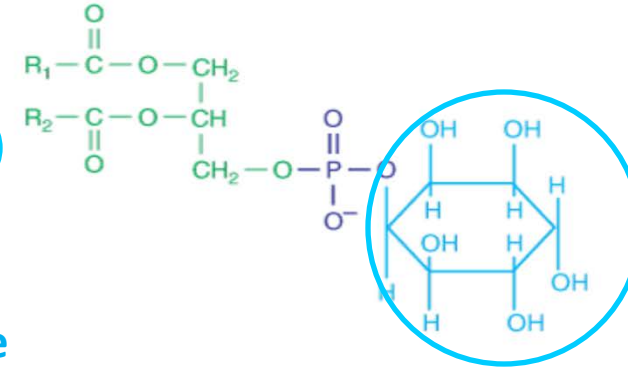
Phosphatidyl**serine**



Phosphatidyl**choline**



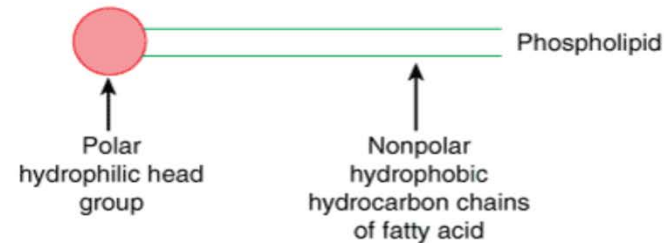
Phosphatidyl**ethanolamine**



Phosphatidyl**inositol**

Glycerophospholipids

From: Chapter 3. Lipids
The Big Picture: Medical
Biochemistry, 2018



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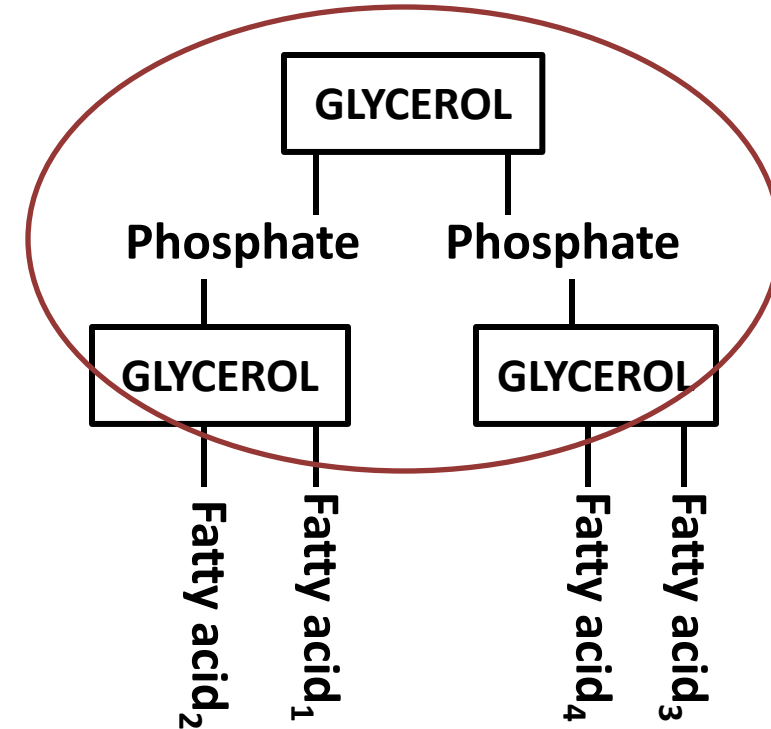
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Common Phospholipids Found in Humans. Common head groups, which form phospholipids (top and middle) and are shown in blue, bonded to the hydroxyl (OH) of a phosphate group (purple), which itself is bonded to the hydroxyl (OH) of glycerol carbon 3 (green). This structure results in a phospholipid molecule that is generally found in membranes. Lipid molecules are often represented in “stick diagram” form (bottom) with the charged phosphate group and hydrophilic head group shown as a circle and oval and with the fatty acid “tails,” depicted as either straight or jagged lines, forming the hydrophobic region.

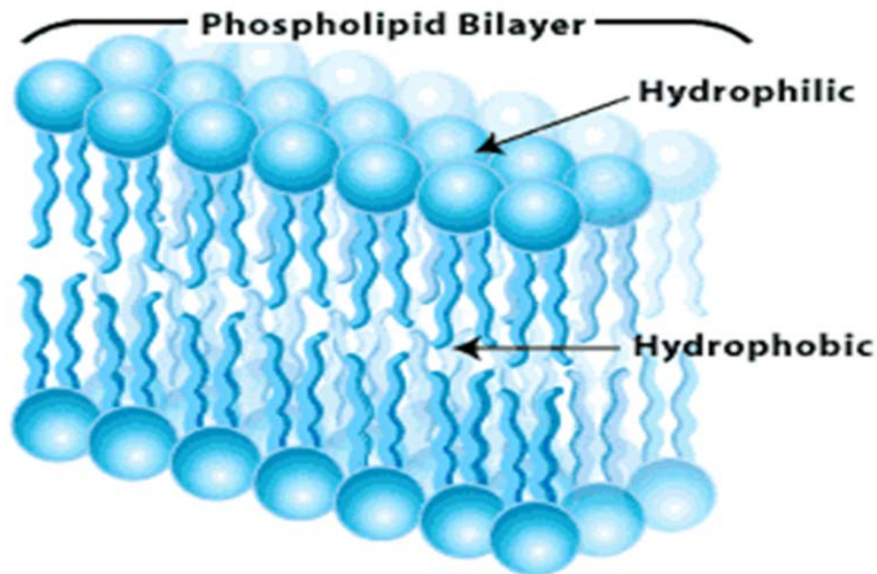
Glycerophospholipids: Examples

- Cardiolipin (diphosphatidylglycerol)
 - Only in **inner mitochondrial membrane**
 - **Anticardiolipin antibodies:** Autoimmune disorders (SLE)

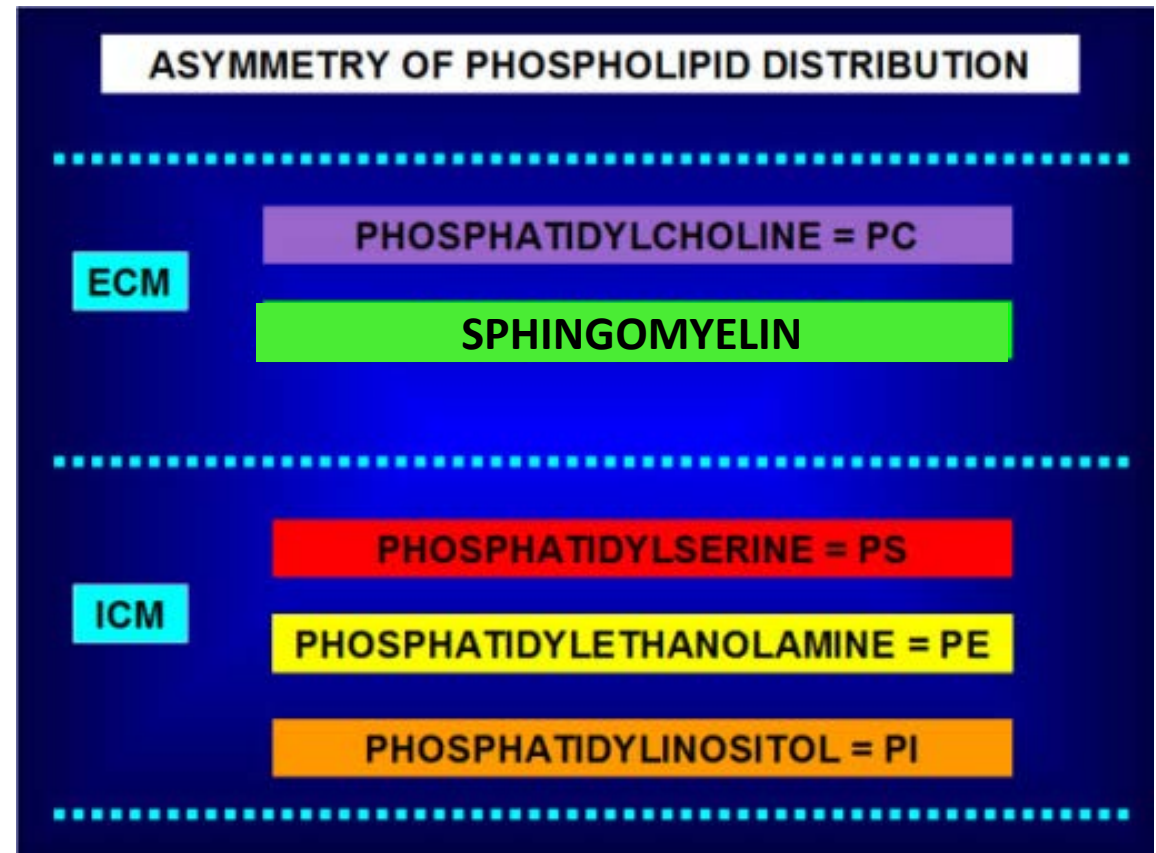


Glycerophospholipids in cell/plasma membrane

Extracellular/ outer leaflet (ECM)



Intracellular/ inner leaflet (ICM)

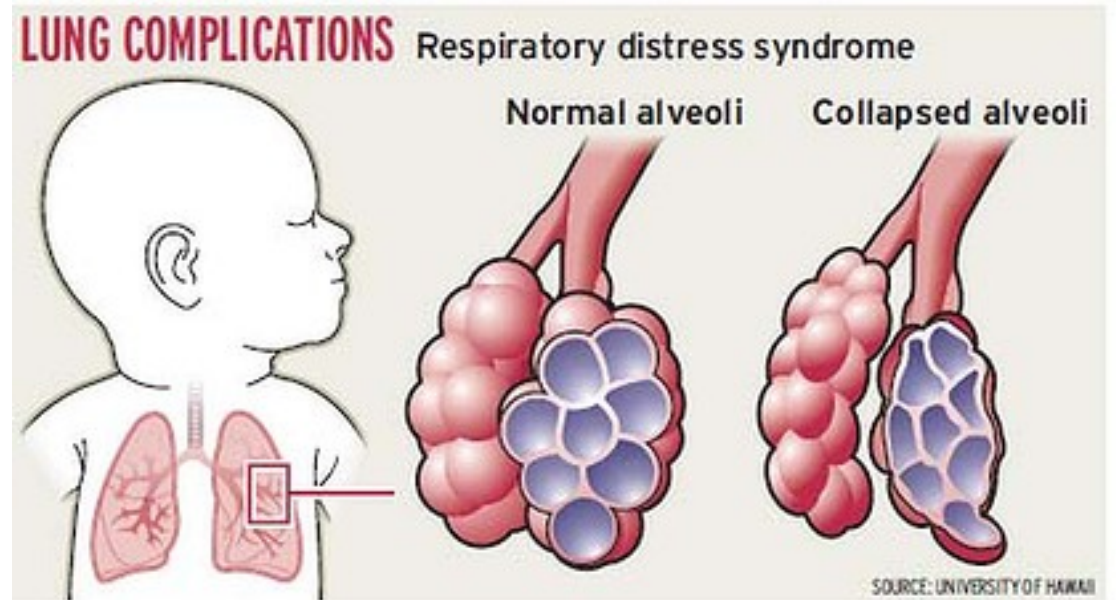
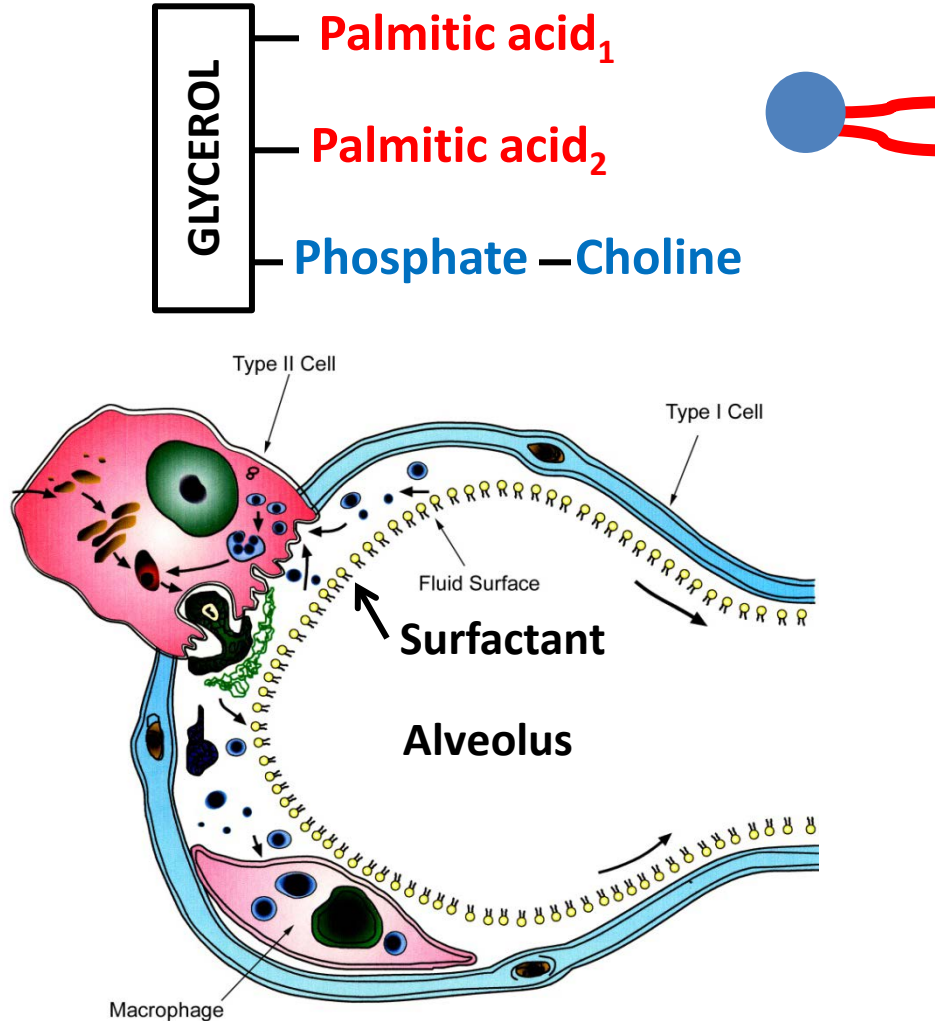


Glycerophospholipids in cell/plasma membrane

- **Phosphatidyl choline (lecithin):** Most abundant
 - In both leaflets of membrane bilayer
 - More abundant in **outer leaflet**
- **Phosphatidylethanolamine and phosphatidylserine* (Apoptosis)**
 - **Inner leaflet** of phospholipid bilayer (Cytosolic side)
 - Refer DLA on Membrane structure
- **Phosphatidylinositol** and phosphatidylinositol bisphosphate (PIP2)
 - Inner leaflet of bilayer
 - Second messengers: **Inositol trisphosphate (IP_3)** and **diacylglycerol (DAG)**

Phospholipid in surfactant

Dipalmitoylphosphatidylcholine; Lecithin; Dipalmitoyl lecithin

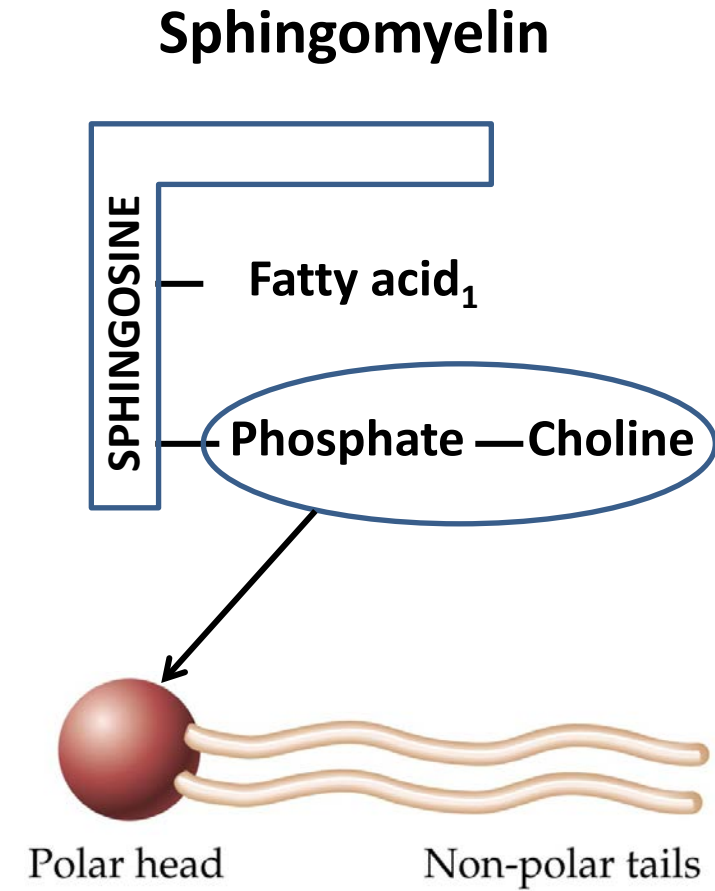


Dipalmitoylphosphatidylcholine (DPPC): Lung surfactant

- Decreases surface tension of alveolar fluid
 - Prevents alveolar collapse
- Respiratory distress syndrome: Collapse of alveoli
 - Premature infants
 - Chronic smokers (**Smoking destroys type II cells**)
- Amniotic fluid Lecithin-Sphingomyelin ratio (L/S ratio)
- **L/S ratio > 2**: Mature fetal lung and low risk of RDS
- **L/S ratio < 1.5**: Immature fetal lung
- Artificial surfactant fluids with DPPC/ Maternal steroid injections
- **What group of lipid does DPPC belong to? Glycerophospholipid**

Sphingophospholipid (sphingomyelin)

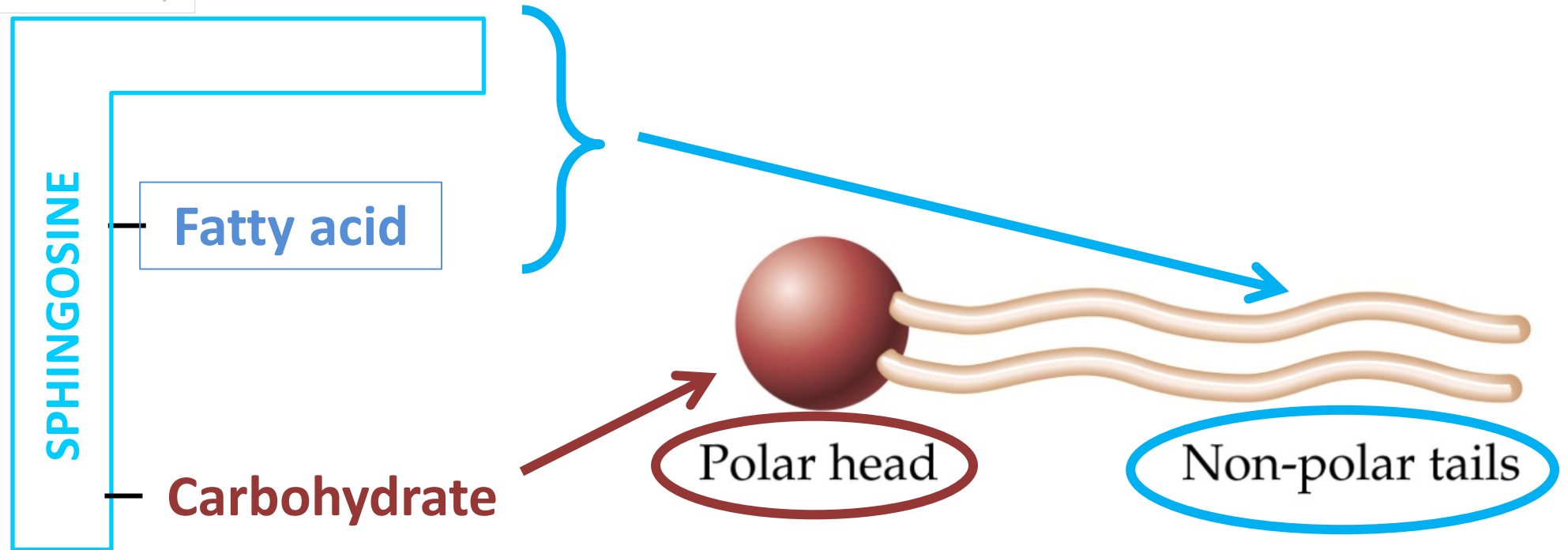
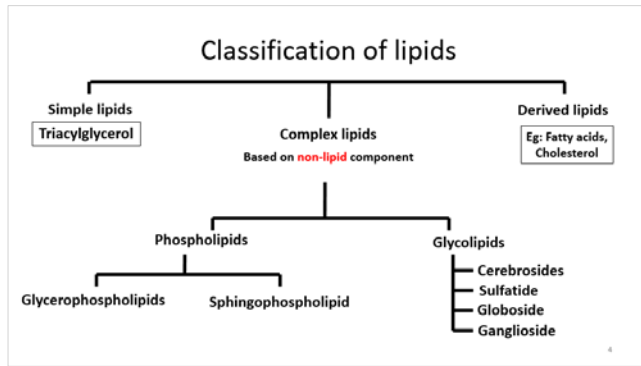
- Sphingosine (C-18)
- Sphingosine with fatty acid forms '**ceramide**'
- **Sphingosine - fatty acid – Phosphate – Choline**
- **Myelin sheath** around neurons



Phospholipids – Composition and function

Lipid	Composition	Function/ Significance
Phosphatidylcholine (Lecithin)	Glycerol + 2 Fatty acids + Phosphate + Choline	Outer leaflet of cell membrane. DPPC: lung surfactant
Phosphatidylserine	Glycerol + 2 Fatty acids + Phosphate + Serine	Inner leaflet of cell membrane; Presence in outer leaflet: Apoptosis
Phosphatidylinositol	Glycerol + 2 Fatty acids + Phosphate + Inositol	Inner leaflet of cell membrane; Forms second messengers
Phosphatidylethanolamine	Glycerol + 2 Fatty acids + Phosphate + Ethanolamine	Inner leaflet of cell membrane
Cardiolipin	--	Inner mitochondrial membrane
Sphingomyelin	Sphingosine + Fatty acid + Phosphate + Choline	Outer leaflet of plasma (cell) membrane

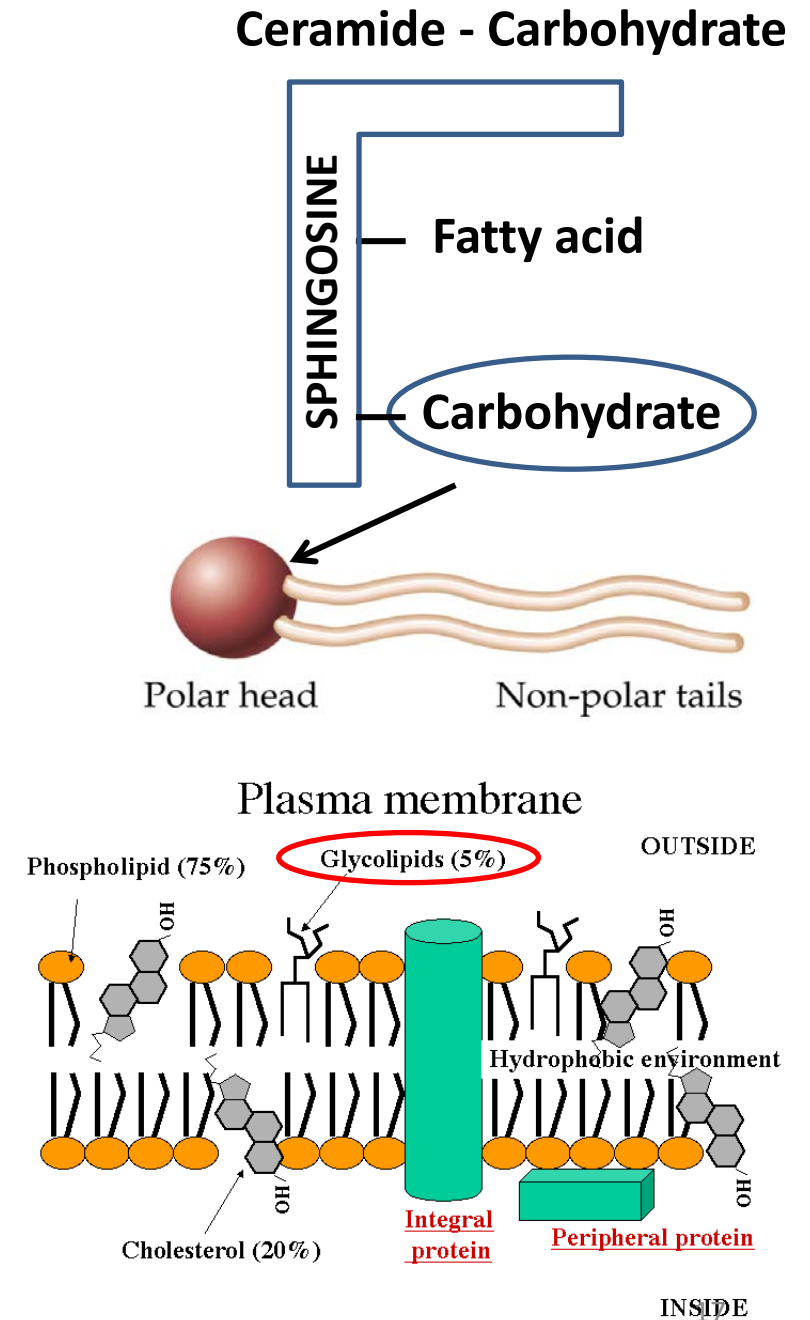
Glycolipids



Ceramide - Carbohydrate

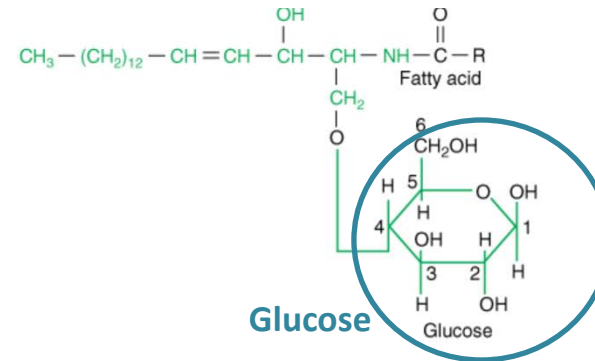
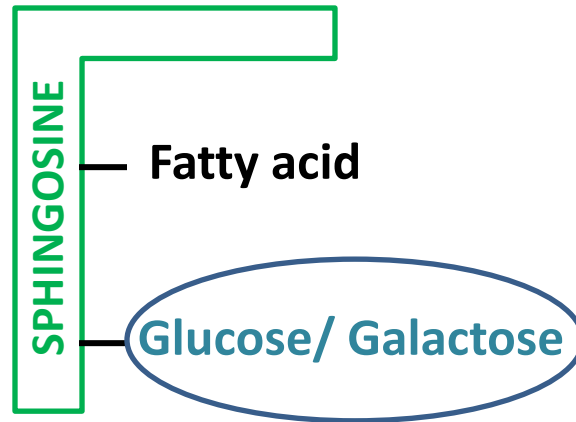
Glycolipids

- **Amphipathic**
- Sphingosine and fatty acid: Nonpolar tails
- Carbohydrate: Polar head group
- **Outer leaflet** of plasma/ cell membrane
- **Glycocalyx** with glycoproteins on **outer leaflet** of cell membrane
- Function as receptors
- Myelin sheath of nerves

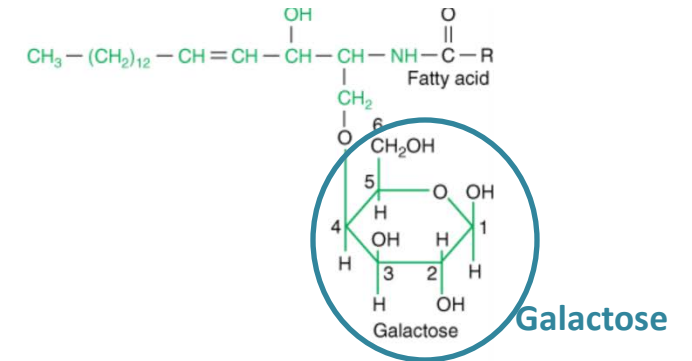


Cerebrosides and Sulfatide

Cerebrosides

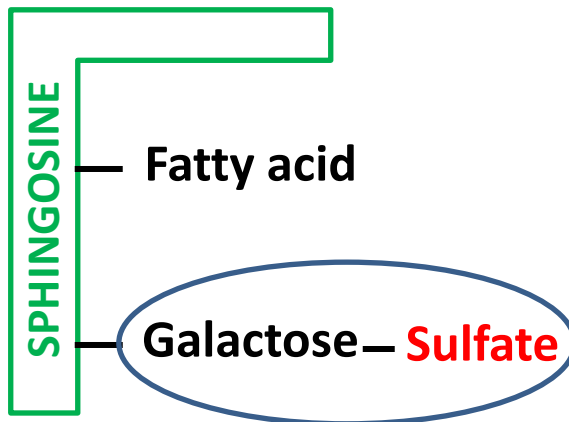


Glucosylceramide



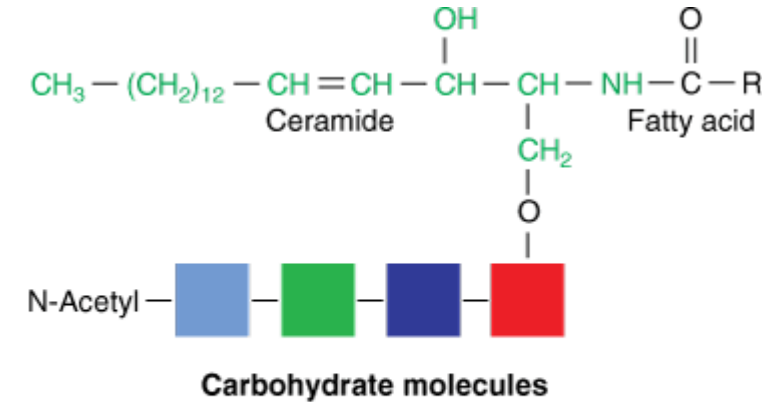
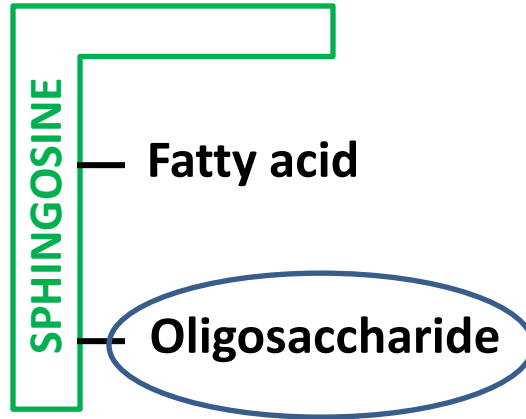
Galactosylceramide

Sulfatide



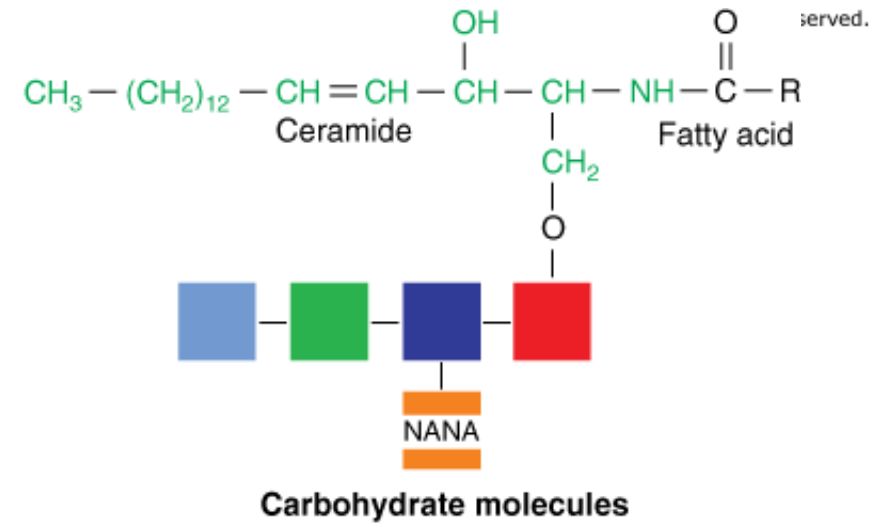
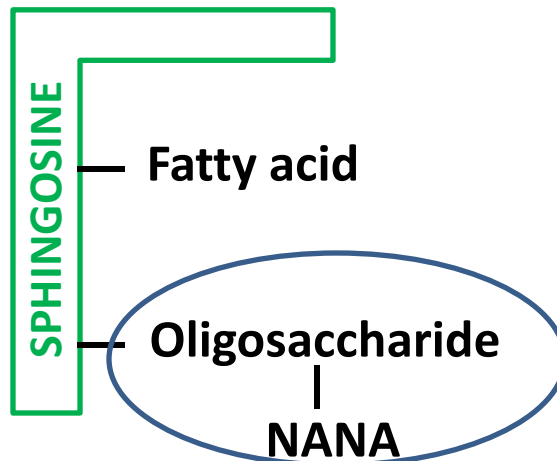
Globosides and Gangliosides

Globosides



Source: Janson LW, Tischler ME: *The Big Picture: Medical Biochemistry*; www.accessmedicine.com

Gangliosides (GM)



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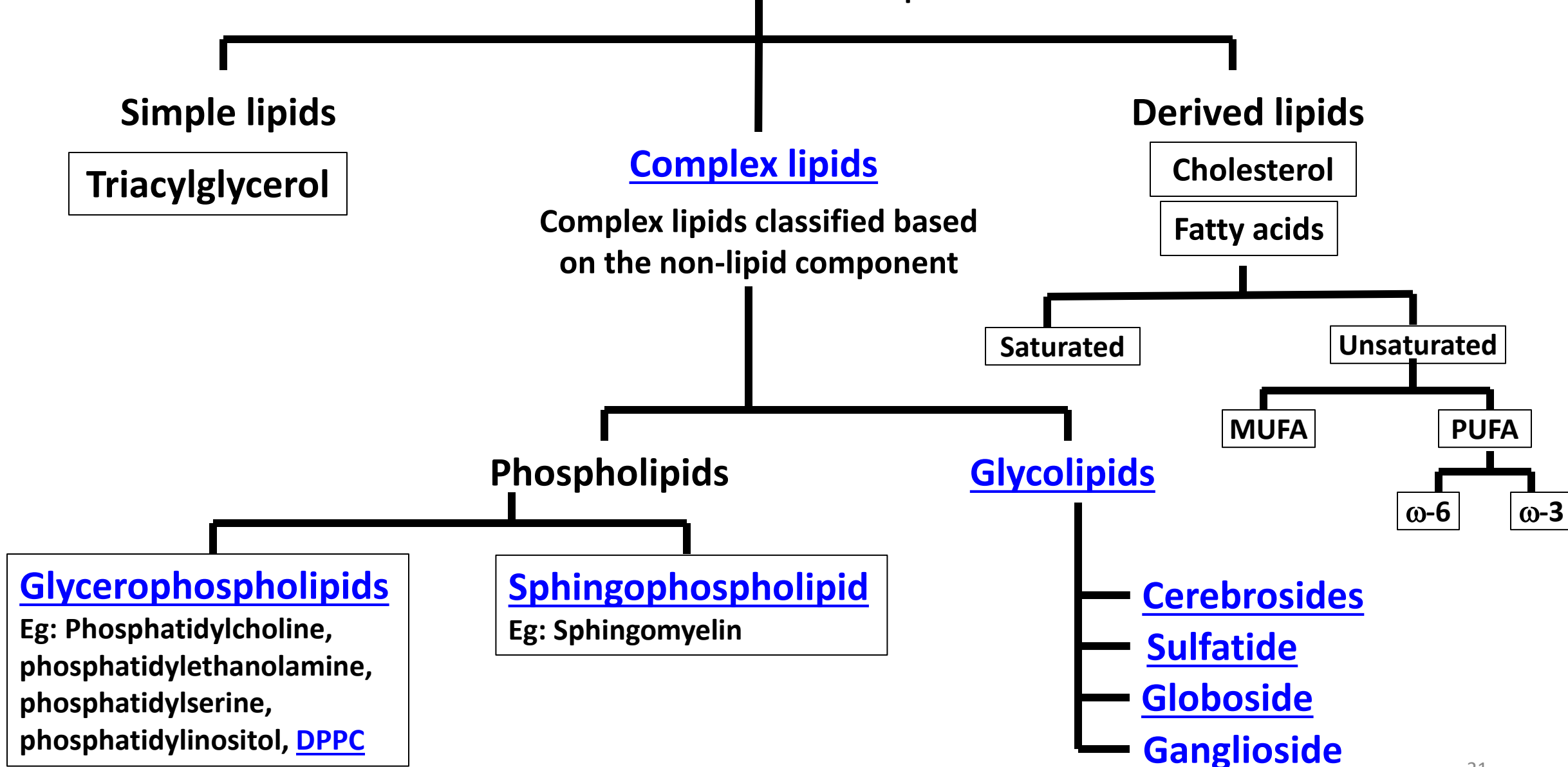
NANA: N-acetyl neuraminic acid (9C sugar)

Glycolipids – Composition and function

Lipid	Composition	Function/ Significance
Cerebroside	Sphingosine + Fatty acid + Monosaccharide	Outer leaflet of plasma (cell) membrane Forms glycocalyx
Sulfatide	Sphingosine + Fatty acid + Monosaccharide + Sulfate	
Globoside	Sphingosine + Fatty acid + Oligosaccharide	
Ganglioside	Sphingosine + Fatty acid + Oligosaccharide (with NANA) Tay-Sachs disease	

Summary

Classification of lipids





Nutritional/ Medical significance

Membrane lipids

Thank You!!

Please email shupadhya@sgu.edu for questions